

1 Introduction

This sprint, like the last one, was spent entirely on fixing and optimizing anything that we might have missed since the start of the challenge. Essentially, we read almost the entire codebase and took notes on what we didn't like and then we fixed everything that caught our attention. This included everything from small changes and fixes to refactoring and completely rewriting parts of code. Some of our optimizations ultimately failed or slowed down other parts of the system, but we kept our heads high and moved on to other issues. It was a bit weird for us not to add any new features, but make sure everything works optimally instead without any massive, easily noticeable changes.

2 Planned activities

During this sprint, we had a plan to accomplish the following tasks:

- Make sure everything works with new calibrations, remove bugs
- Fix dotted line following
- Implement localization
- Keep optimizing everything

3 Status of planned activities

- **Make sure everything works with new calibrations, remove bugs**

We spent the beginning of this sprint fixing our calculations and making all the tiny changes necessary for the car to behave normally. We even borrowed some books from our professors to help us calculate new PID values. The car now drives and turns smoothly and doesn't make sudden harsh adjustments without proper cause.

- **Fix dotted line following**

A big part of our optimization attempts were focused on lane following and image pre-processing. While our pre-processing attempts ultimately weren't successful, we managed to write a safer line following algorithm that should be perfectly fine with following dotted lines and even driving without seeing any lines for a short period of time.

- **Implement localization**

After much thought and many, many ideas we initially tried to implement localization using the cars camera and an algorithm to compare what the camera sees with the image of the map to try and find where the car is. While this was a good concept, we couldn't quite make it work reliably, so we opted for the simplest approach we thought of. Right now, the car uses path planning information to calculate the distance between important nodes and the speeding info from the STM to get a rough estimate of where it is. We plan to combine this with the localization server to make a complete module. This module is finished but still untested, so it isn't shown in the video report.

- **Keep optimizing everything**

This is a very broad bullet point that covers pretty much everything we've done this sprint, so to keep this report sensibly short we will cover everything in the next section.

4 General status of the project

The car runs much smoother and more reliable than ever before. We paid special attention to the most crucial parts of the system that could hinder overall performance. Our biggest change was rewriting autoFSM from the ground up along with refactoring the lane detection and control modules. We also moved the PIDController module to a separate file for better readability. We added a better roundabout control algorithm and improved lane following and detection logic. Also, we updated our roundabout test track and fixed the path planning code to keep up, along with new intersection control fixes. Another important set of fixes was adjusting the position variables for object detection, along with improving the accuracy and overall reliability of the object detection module. Again, for better readability, we separated sign detection from object detection and added the TrafficSignsDetection message type. Apart from that, we tuned and adjusted a bunch of parameters ranging from intersection control and general driving, to more specific scenarios like highway maneuvers. We are satisfied with our work and are excited to add the finishing touches before the competition days.

5 Upcoming activities

In the next sprint we want to do the following tasks:

- Test localization and complete with server info
- Implement tunnel handling
- Implement fog handling
- Make sure car drives well with no road markings
- Figure out edge cases with unmarked intersections
- Implement reactions to other (active) cars in traffic